

Characteristics Associated with Severe and Fatal Motor Vehicle Crash Outcomes in the United States, 2022–2024

This report examines which behavioral and environmental characteristics are associated with severe or fatal outcomes in U.S. police-reported crashes from 2022 to 2024. It uses weighted data from NHTSA's Crash Report Sampling System (CRSS), descriptive analysis and logistic regression.

1. Introduction

a. Research question

Preventing crashes and reducing the severity of crashes that do occur are related but different safety goals. Only about 3% of U.S. police-reported crashes result in a death or serious injury, but these roughly 194,000 crashes per year carry the greatest human cost. Reducing them requires understanding not just how often crashes happen, but what makes some crashes severe while most are not. That is the question this project addresses:

Which behavioral and environmental characteristics are most strongly associated with severe or fatal outcomes in U.S. police-reported crashes from 2022 to 2024?

Instead of studying one group of drivers, the project looks at several factors together. These are two driver behaviors (speeding and alcohol), when the crash happened (night, weekend), and where it happened (lighting, weather, urban or rural setting, junction type, interstate). Examining them in one model shows which associations remain once the others are held constant.

b. Disciplinary motivation

This project comes from my focus on transportation, traffic and safety analysis. In this field, the main measure of roadway safety is not how many crashes happen but how many people are killed or seriously injured. Transportation agencies classify crash outcomes on the KABCO injury scale. Fatal (K) and suspected serious injury (A) crashes, together called "KA" crashes, are the standard focus of safety planning. Federal safety performance measures for state DOTs are built on counts and rates of fatalities and serious injuries. The U.S. DOT's National Roadway Safety Strategy (2022) adopts the Safe System Approach, whose central principle is that crashes will still happen, but they should not result in death or serious injury.

This shifts the analytical question. Many safety studies ask where and when crashes are frequent. A Safe System perspective also asks which conditions turn a crash into a severe one. The two questions can have different answers. As this analysis shows, the afternoon rush hour has the most crashes, but the hours after midnight have the most severe ones. Similarly, rain and snow increase crash frequency but are associated with less severe outcomes.

The project also applies core data-science skills to a real transportation-safety dataset. These skills are combining multiple years and files, aggregating vehicle-level records to the crash level, using survey weights from a complex probability sample, and building an interpretable statistical model step by step.

2. Data

The data are the 2022, 2023 and 2024 releases of NHTSA's Crash Report Sampling System (CRSS). CRSS is a nationally representative probability sample of police-reported crashes of every severity, from property damage only to fatal. Two files were used for each year:

File	Level	Records 2022–2024	Used for
Accident	One row per crash	155,716	Outcome, alcohol, time, lighting, weather, urban/rural, junction, interstate, sampling weight
Vehicle	One row per vehicle	272,858	Speeding (aggregated to the crash)

The unit of analysis is the crash. `CASENUM` is unique only within a year, so vehicle records were linked to crashes on year and `CASENUM` together. There are no duplicate crash IDs.

Sampling weights. CRSS deliberately oversamples severe crashes, so every crash carries a sampling weight (`WEIGHT`). In the raw sample, 13.2% of crashes are severe or fatal; once the weights are applied, the national share is 3.2%. All percentages in this report are therefore weighted. With three years pooled, $WEIGHT / 3$ gives average crashes per year. The weighted total is about 6.1 million police-reported crashes per year.

3. Data cleaning and variables

A crash is coded severe/fatal = 1 if its most severe injury was fatal (K) or a suspected serious injury (A), and 0 if it was a suspected minor injury (B), possible injury (C) or no apparent injury (O). The variable is `MAXSEV_IMNAME`, NHTSA's imputed maximum severity. 408 crashes (0.26%) were excluded because their severity could not be placed on this scale. The analytic file has 155,308 crashes.

The CRSS label columns (`...NAME`) were used directly and grouped into a small set of readable categories:

Group	Variable	Categories (reference first)
Behavioral	Speeding	No, Yes, Unknown
Behavioral	Alcohol	No, Yes (NHTSA-imputed <code>ALCHL_IMNAME</code>)
Time	Night	Day/evening, Night (21:00–05:59)
Time	Weekend	Weekday, Weekend
Environment	Lighting	Daylight, Dawn/Dusk, Dark-lighted, Dark-not lighted,

Group	Variable	Categories (reference first)
		Dark-unknown (EDA only)
Environment	Weather	Clear, Cloudy, Rain, Snow/ice, Other
Roadway	Urban/rural	Urban, Rural
Roadway	Junction	Non-junction, Intersection, Driveway, Interchange/ramp, Other
Roadway	Interstate	No, Yes, Unknown

Speeding from vehicle to crash. Speeding is recorded per vehicle, so it was aggregated to the crash. A crash is coded *Yes* if at least one vehicle was speeding. It is coded *Unknown* if no vehicle was known to be speeding but at least one had unknown status. Otherwise it is *No*. This gives 16,118 speeding crashes, 132,860 without speeding and 6,330 unknown.

Unknown values were kept as their own category in the EDA rather than treated as "No". The regression excludes the 6,342 crashes (4.1%) with unknown speeding or interstate status.

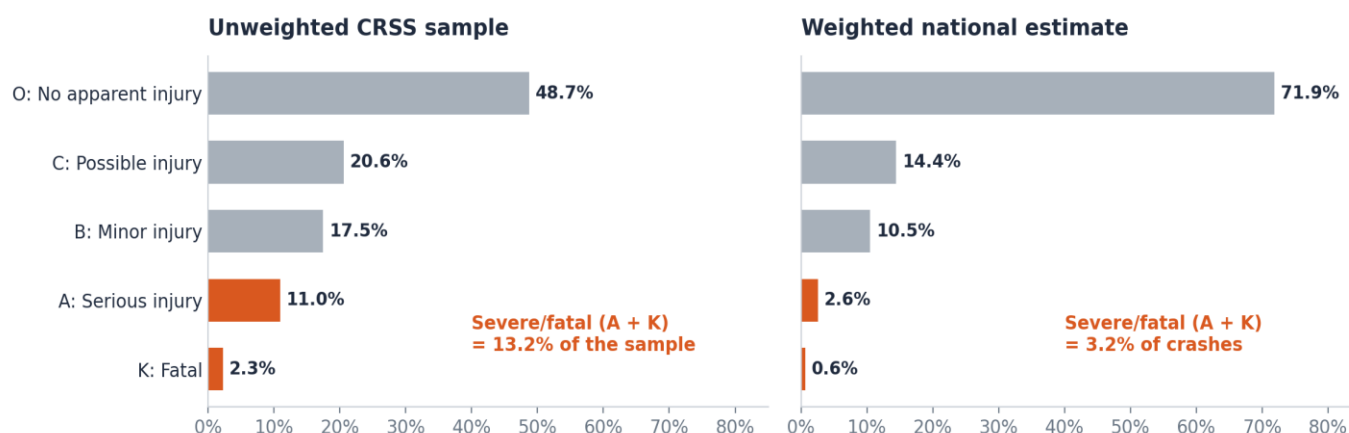
4. Exploratory analysis

All percentages in this section are weighted national estimates. The sample size behind each group is shown on the charts.

a. How severe are the crashes?

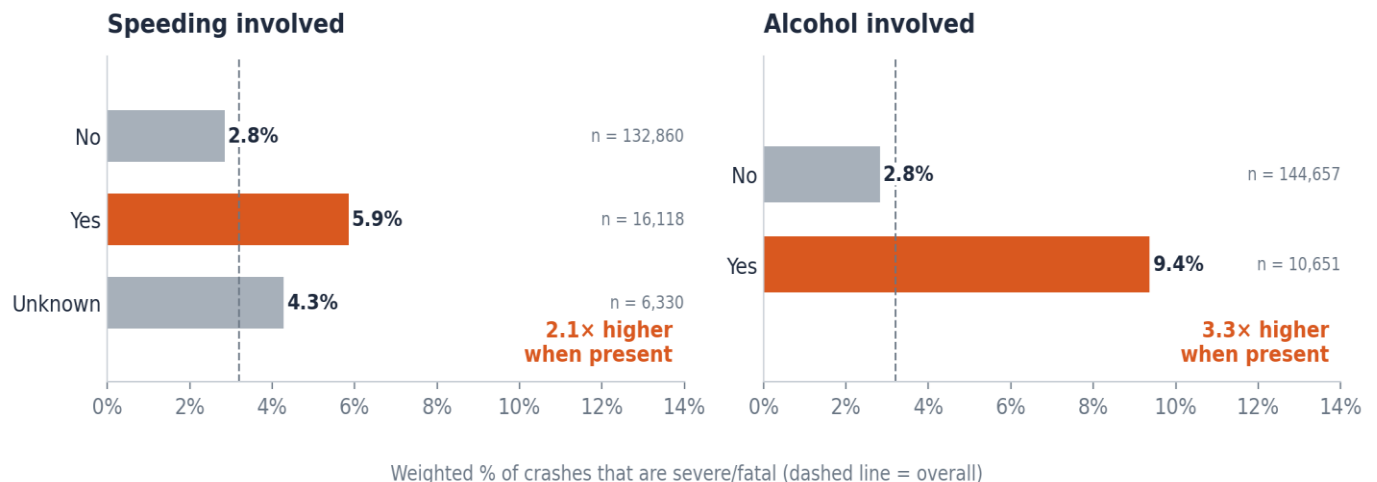
About 3.2% of police-reported crashes are severe or fatal, roughly 194,000 crashes per year. Most crashes (71.9%) cause no apparent injury. In the raw sample the severe/fatal share is four times higher (13.2%), which shows why the weights are needed. The share was stable across the three years: 3.3% in 2022 and 3.1% in both 2023 and 2024.

Crash outcomes by maximum injury severity, 2022-2024



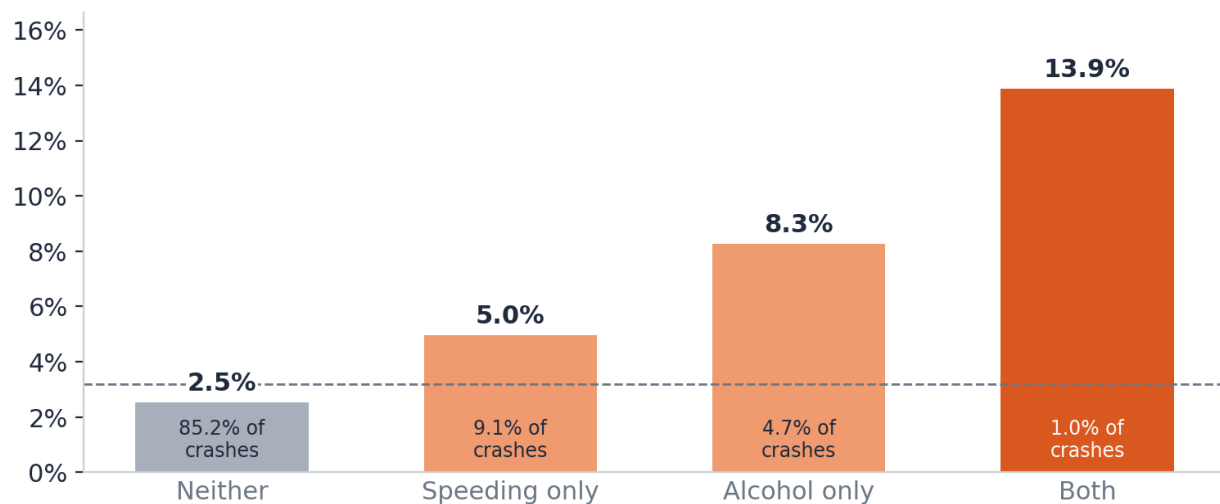
b. Behavior: speeding and alcohol

When alcohol is involved, 9.4% of crashes are severe or fatal, compared with 2.8% without it (3.3 times higher). With speeding, the figures are 5.9% versus 2.8% (2.1 times higher). Crashes with unknown speeding status sit in between, at 4.3%.



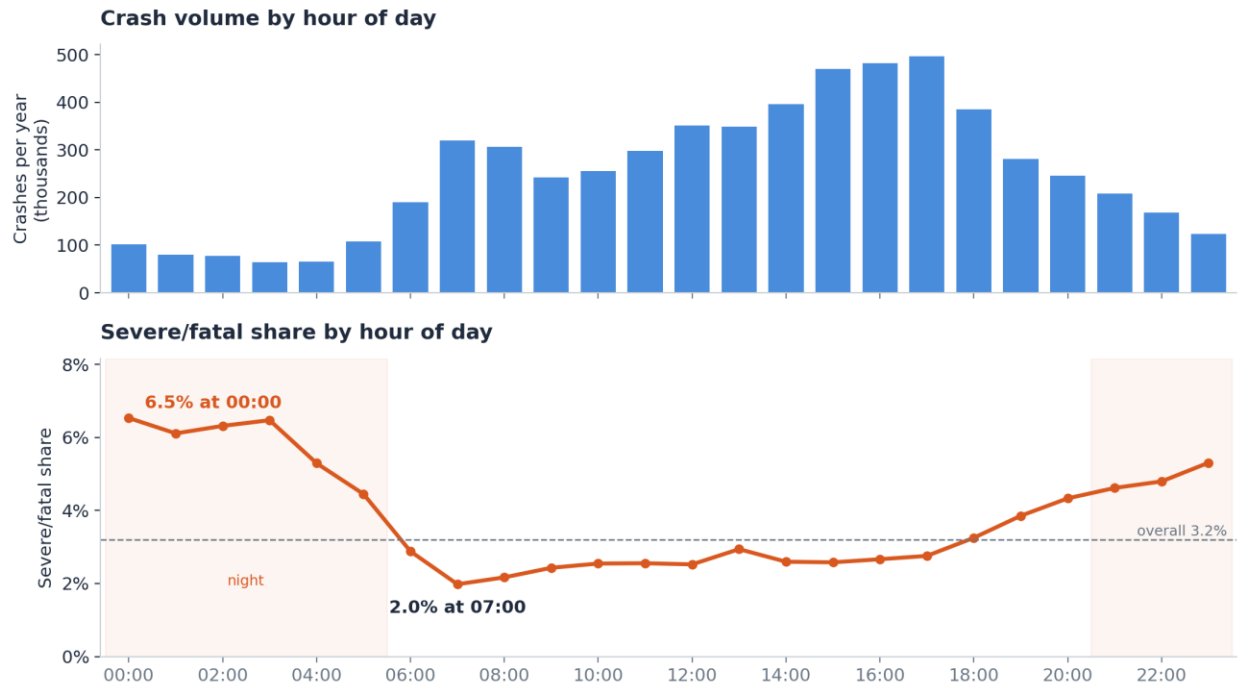
The two behaviors compound each other. Crashes with both speeding and alcohol are only about 1% of crashes, but 13.9% of them are severe or fatal. That is more than five times the 2.5% for crashes with neither.

Severe/fatal share by combination of speeding and alcohol

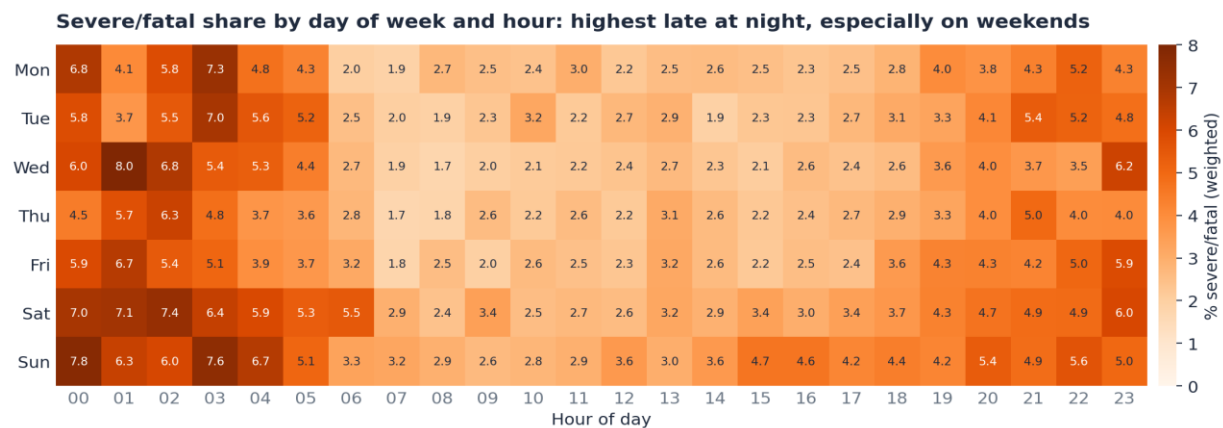


c. Time of day and day of week

Crash volume peaks in the weekday afternoon rush between 15:00 and 18:00. Crash severity does the opposite, it is highest between midnight and 04:00 (6.5% at 00:00) and lowest in the morning commute (2.0% at 07:00). Night crashes (21:00–05:59) are almost twice as likely to be severe as day/evening crashes (5.3% vs 2.8%). Weekend crashes are also more severe (4.2% vs 2.9%).



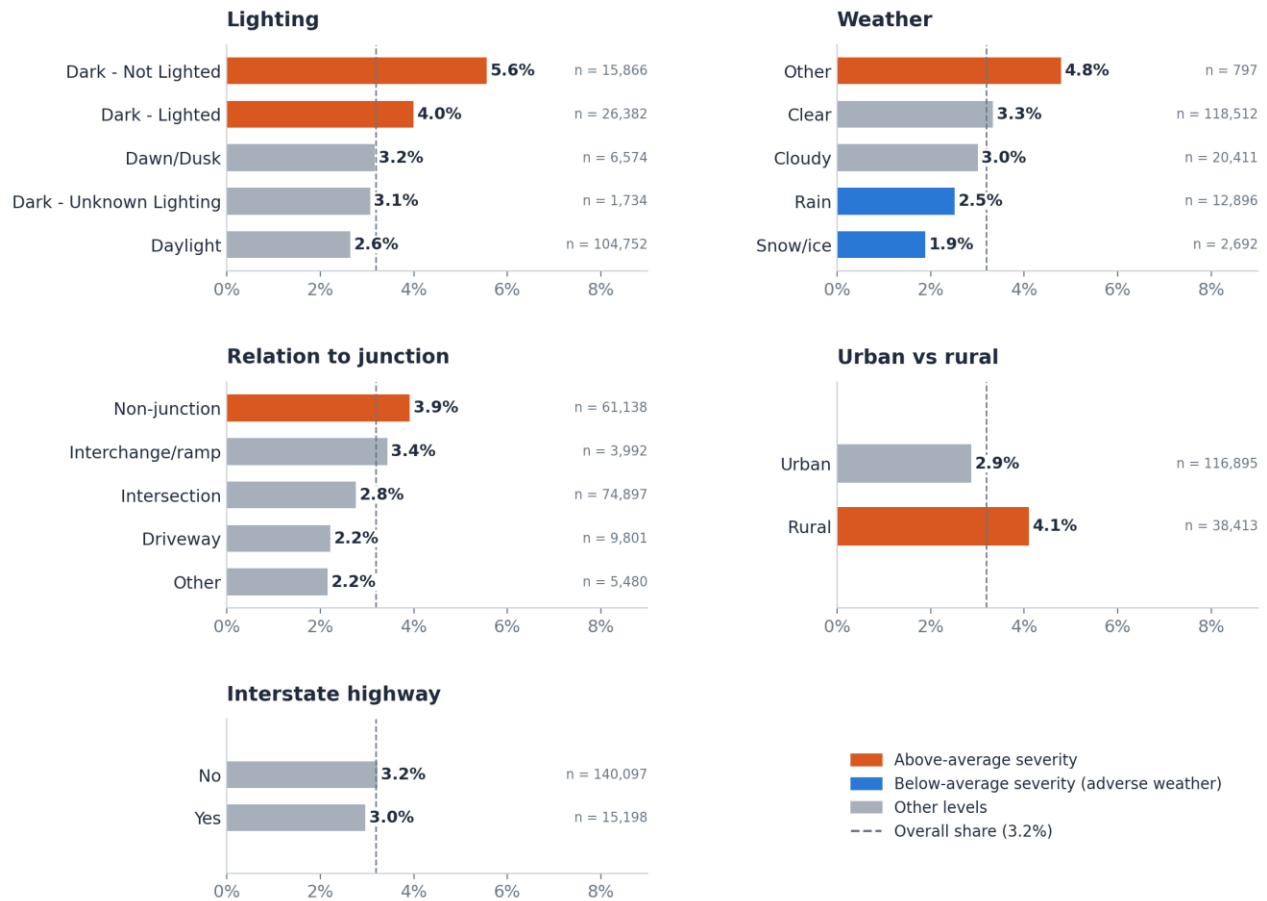
The day × hour heatmap shows that the most dangerous hours are Friday-to-Sunday nights.



d. Environment and roadway

- **Lighting:** Severity rises from daylight (2.6%) to dark but lighted roads (4.0%) and dark, unlighted roads (5.6%).
- **Weather:** Adverse weather runs against intuition: rain (2.5%) and snow/ice (1.9%) crashes are *less* often severe than clear-weather crashes (3.3%). A plausible explanation is that drivers slow down in bad conditions. The small "other" group (fog, wind and similar; n = 797) is the exception at 4.8%.
- **Roadway:** Rural crashes (4.1% vs 2.9% urban) and crashes away from junctions (3.9%) are more often severe than intersection (2.8%) or driveway crashes (2.2%). Interstate crashes are not more severe than others (3.0% vs 3.2%).

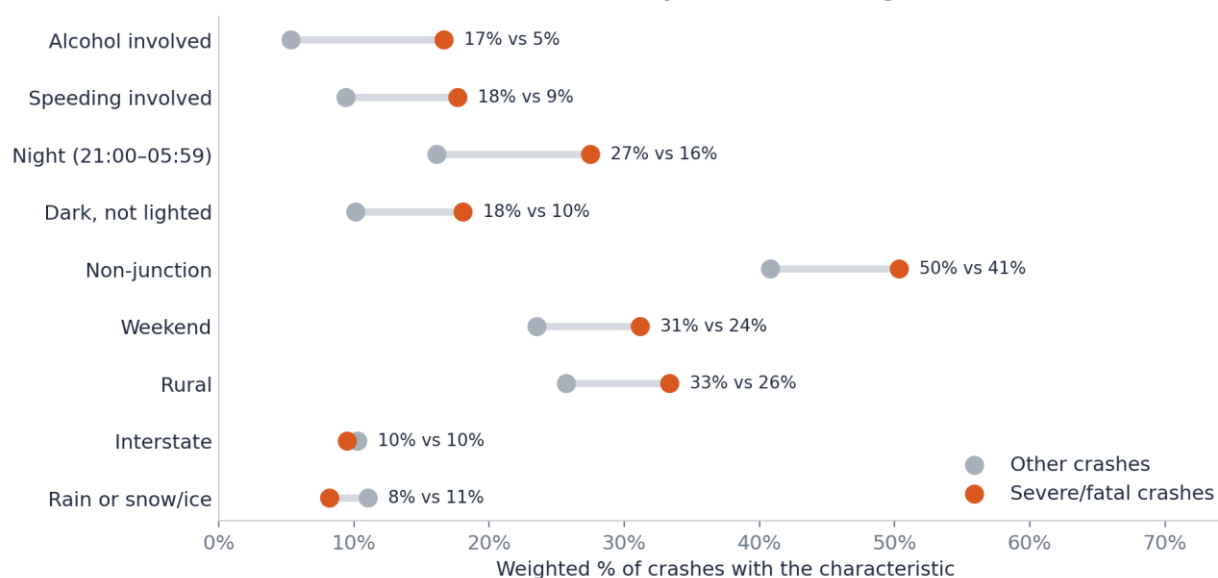
Weighted severe/fatal share by environmental and roadway characteristics



e. Which characteristics stand out?

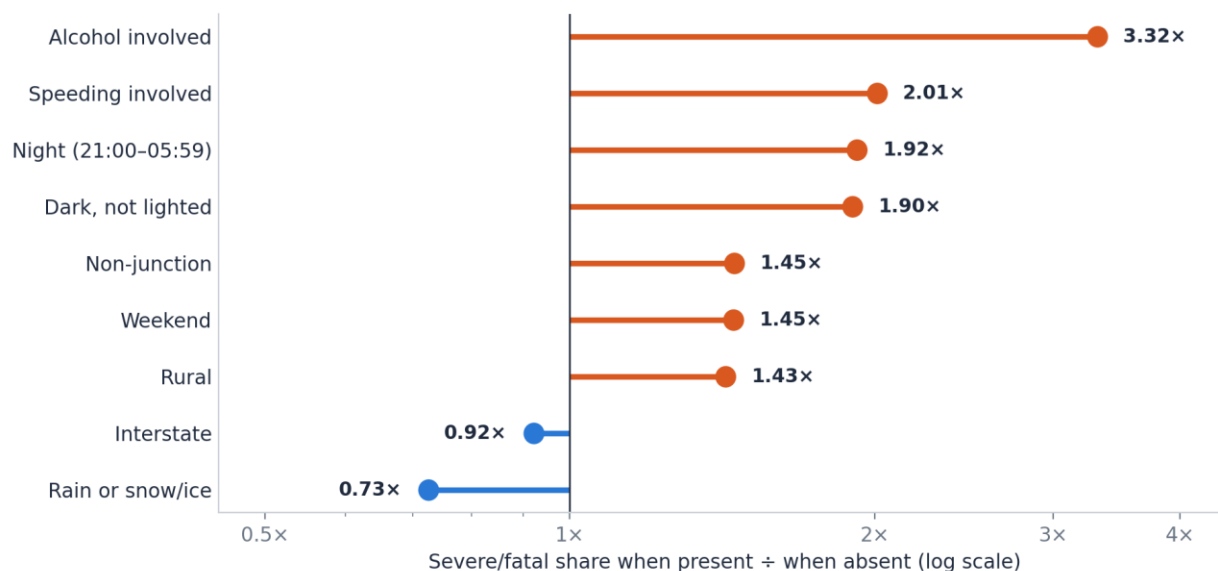
Comparing severe/fatal crashes with all other crashes shows which characteristics are over-represented. For example, alcohol is involved in 17% of severe/fatal crashes but only 5% of other crashes. Speeding appears in 18% versus 9%, and night in 27% versus 16%.

Which characteristics are over-represented among severe/fatal crashes?



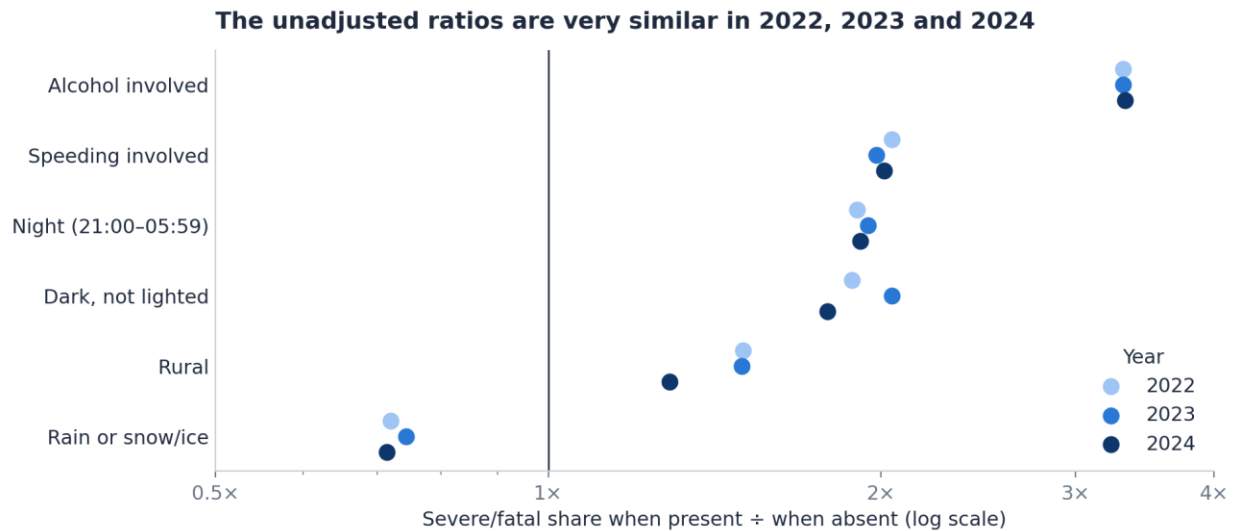
Dividing the severe/fatal share when a characteristic is present by the share when it is absent ranks the characteristics. Alcohol (3.3×), speeding (2.0×), night (1.9×) and dark unlighted roads (1.9×) lead. Non-junction, weekend and rural locations follow at about 1.4–1.5×. Rain or snow/ice (0.73×) is the only characteristic linked to lower severity. This ranking is unadjusted: it ignores the overlap between characteristics, which Section 5 addresses.

Unadjusted ranking: alcohol, speeding and night stand out



f. Stability over time

The unadjusted ratios are nearly identical in each year: alcohol 3.31, 3.31 and 3.33; speeding 2.05, 1.98 and 2.01; night 1.90, 1.95 and 1.92. Only the rural ratio is lower in 2024 (1.29 vs 1.50). This stability supports pooling the three years.



5. Weighted logistic regression

The exploratory analysis looked at one characteristic at a time, but many of them overlap. Alcohol-involved crashes, for instance, are concentrated at night. The regression therefore estimates all characteristics together to identify which ones remain associated with a severe or fatal outcome when the others are held constant.

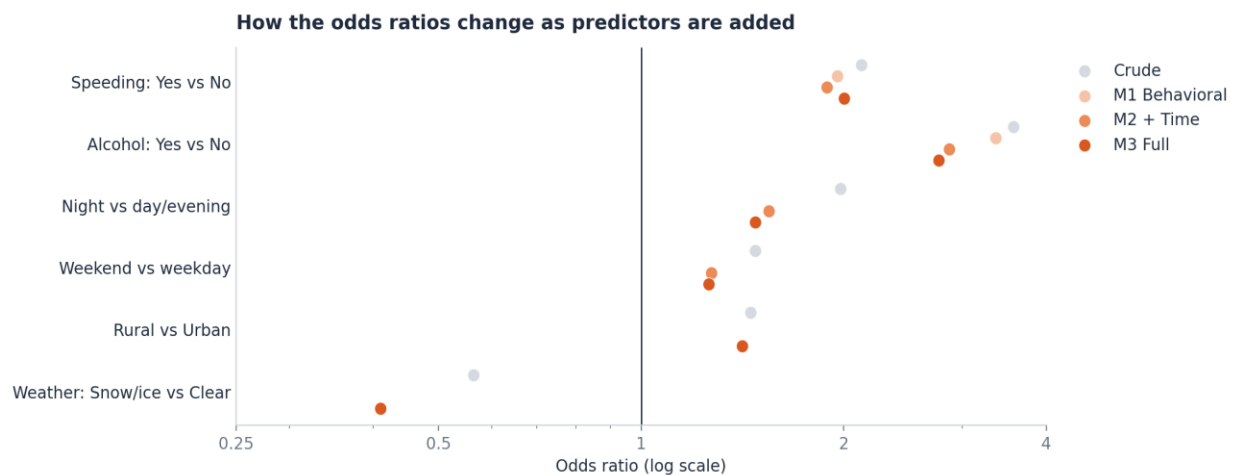
a. Setup

- **Data:** 148,966 crashes, after excluding the 4.1% with unknown speeding or interstate status. Their severe/fatal share (4.3%) is somewhat above average, so the exclusion is noted as a limitation. The sample contains 19,470 severe/fatal crashes, and every category has at least 141 of them, enough to estimate each odds ratio.
- **Model:** A logistic regression of severe/fatal (1/0) on speeding, alcohol, night, weekend, weather, urban/rural, junction type, interstate and year, fitted with `statsmodels glm` (binomial family).
- **Weights:** The CRSS weights, normalized to a mean of 1, are used as `var_weights`. Normalizing does not change the estimates.
- **Standard errors:** Crashes are sampled in geographic clusters (primary sampling units, `PSU_VAR`), and crashes from the same area tend to be similar. Standard errors are therefore cluster-robust by PSU (`cov_type="cluster"`, 67 clusters). This does not change the odds ratios, but it gives more honest confidence intervals than the default.
- **Night vs. lighting:** These overlap almost completely: 95% of night crashes occur in the dark, against 14% of day/evening crashes. Including both would split one signal between two variables, so the model uses `night`, which has no unknown values. `Lighting` stays in the exploratory analysis.
- This is an approximate approach: `statsmodels` flags cluster-robust errors with `var_weights` as "not fully supported", and the sampling strata (`PSUSTRAT`) are not used. Section 5.c therefore checks the final model against design-based standard errors.

b. Building the model in three steps

The model was built in steps to see how each association changes as other characteristics are added:

Model	Predictors
Crude	Each predictor alone
M1 – Behavioral	Speeding, alcohol
M2 – Add Time	M1 + night, weekend
M3 – Full	M2 + weather, urban/rural, junction, interstate, year



- Alcohol and speeding stay large in every model. Alcohol falls from 3.58 (crude) to 2.77, partly because alcohol crashes happen mostly at night. Speeding barely moves (2.12 → 2.00).
- Night shrinks from 1.98 to 1.48. Part of the night pattern reflects who is driving and how at night (alcohol and speeding), but a clear association remains.
- Weekend shrinks from 1.48 to 1.26 once night is added, because more weekend crashes happen at night. A smaller weekend association remains.
- Rural (1.45 → 1.41) is almost unchanged. Snow/ice becomes even more protective (0.56 → 0.41).
- McFadden's pseudo- R^2 rises from 0.025 (M1) to 0.030 (M2) and 0.037 (M3). Such low values are typical when a rare outcome is modeled from a few crash-level characteristics. The model describes associations; it is not meant to predict individual crashes.

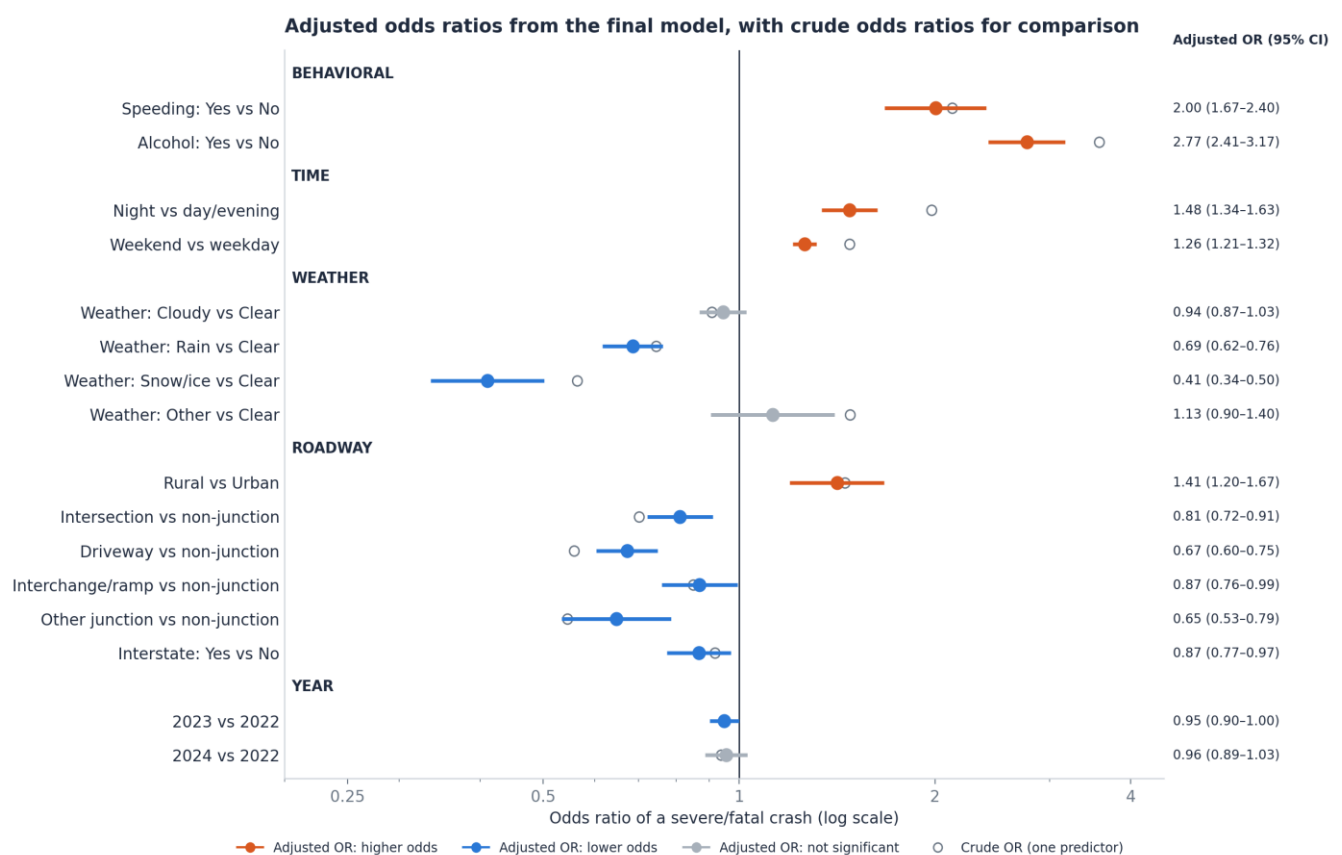
c. Final model: adjusted odds ratios

Comparison	Crude OR	Adjusted OR (95% CI)	Change in odds	p
Alcohol: yes vs no	3.58	2.77 (2.41–3.17)	+177%	< 0.001
Speeding: yes vs no	2.12	2.00 (1.67–2.40)	+100%	< 0.001
Night vs day/evening	1.98	1.48 (1.34–1.63)	+48%	< 0.001
Rural vs urban	1.45	1.41 (1.20–1.67)	+41%	< 0.001
Weekend vs weekday	1.48	1.26 (1.21–1.32)	+26%	< 0.001
Weather: cloudy vs clear	0.91	0.94 (0.87–1.03)	–6%	0.18
Weather: rain vs clear	0.75	0.69 (0.62–0.76)	–31%	< 0.001
Weather: snow/ice vs clear	0.56	0.41 (0.34–0.50)	–59%	< 0.001
Weather: other vs clear	1.48	1.13 (0.90–1.40)	+13%	0.29
Intersection vs non-junction	0.70	0.81 (0.72–0.91)	–19%	< 0.001
Driveway vs non-junction	0.56	0.67 (0.60–0.75)	–33%	< 0.001
Interchange/ram p vs non-junction	0.85	0.87 (0.76–0.99)*	–13%	0.04*
Other junction vs non-junction	0.54	0.65 (0.53–0.79)	–35%	< 0.001
Interstate: yes vs no	0.92	0.87 (0.77–0.97)	–13%	0.01
2023 vs 2022	0.95	0.95 (0.90–1.00)	–5%	0.05
2024 vs 2022	0.94	0.96 (0.89–1.03)	–4%	0.24

Bold = 95% CI excludes 1. "Change in odds" = $(OR - 1) \times 100\%$.

** Not significant under the design-based check ($p = 0.063$). The 2023 vs 2022 difference is also not significant under that check ($p = 0.075$).*

Checking the standard errors against the CRSS design. Because the main model's standard errors ignore the sampling strata, the final model's standard errors were recomputed by Taylor-series linearization using both strata (PSUSTRAT) and primary sampling units (PSU_VAR), with a t distribution on 42 design degrees of freedom. The design-based standard errors were within about $\pm 20\%$ of the clustered ones. All main associations remained significant: alcohol, speeding, night, weekend, rural, rain, snow/ice, intersection, driveway and interstate. Two borderline results did not: interchange/ramp ($p = 0.041 \rightarrow 0.063$) and 2023 vs 2022 ($p = 0.049 \rightarrow 0.075$). These are treated as showing no clear difference. Because the p-values are approximate, results close to the 0.05 threshold are not emphasized.



Interpretation, holding the other characteristics constant:

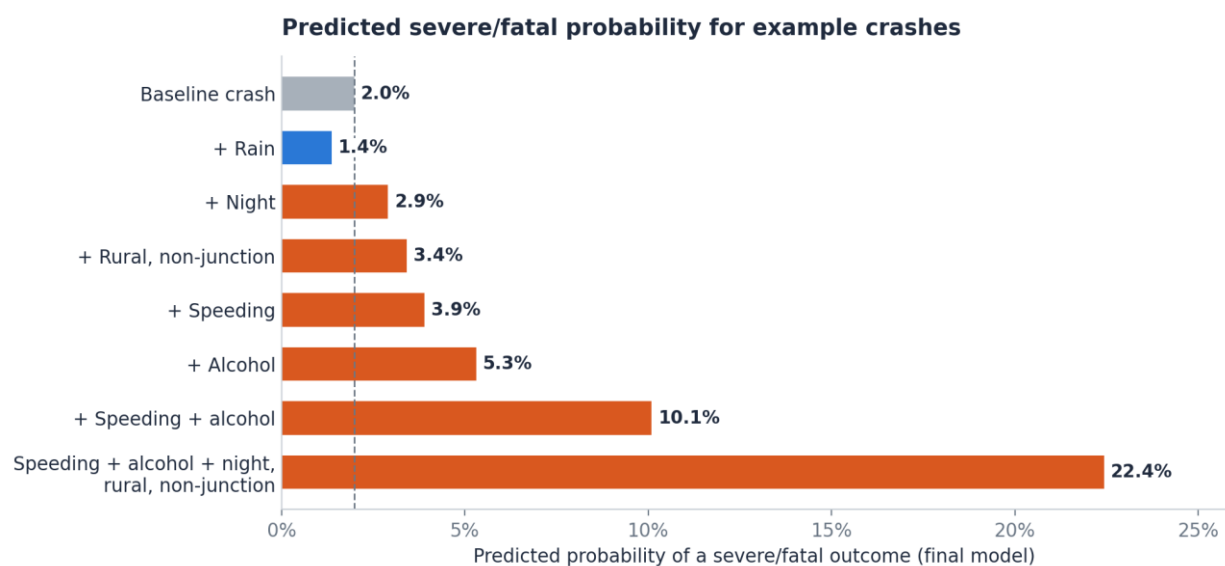
- **Alcohol** is the strongest association: alcohol-involved crashes have about 2.8 times the odds of a severe/fatal outcome.
- **Speeding** doubles the odds.
- **Night** crashes have about 1.5 times the odds of day/evening crashes, even after accounting for alcohol and speeding.
- **Rural** crashes have about 1.4 times the odds of urban crashes, and weekend crashes about 1.3 times the odds of weekday crashes.

- **Rain** (0.69) and **snow/ice** (0.41) are associated with lower odds than clear weather, consistent with drivers slowing down in poor conditions.
- Crashes at **intersections, driveways and other junctions** have lower odds than non-junction crashes. **Interstate** crashes have somewhat lower odds (0.87).
- **Year** shows no clear trend.

Because severe/fatal crashes are rare (about 3%), odds ratios are close to risk ratios here, so an OR of 2 roughly means "about twice as likely".

d. Predicted probabilities for example crashes

To make the odds ratios concrete, the final model predicts the probability of a severe/fatal outcome for a few example crashes. The baseline is a 2024 weekday, day/evening crash in clear weather at an urban intersection, not on an interstate, with no speeding or alcohol.



- The baseline crash has a predicted probability of 2.0%, and rain lowers it to 1.4%.
- Adding one high-risk characteristic raises it to 2.9% (night), 3.4% (rural, non-junction), 3.9% (speeding) or 5.3% (alcohol).
- Speeding plus alcohol gives 10.1%. Adding night and a rural, non-junction location on top of that gives 22.4%, about 11 times the baseline.

These are model predictions for illustration, not forecasts for individual crashes.

6. Discussion, limitations, and conclusion

a. What the results mean

- **Impaired and speeding drivers are the clearest behavioral targets.** Both keep large associations after adjustment (2.77 and 2.00), they compound each other, and their patterns did not change across the three years. This supports keeping impaired-driving and speed enforcement at the center of injury-reduction efforts.

- **Night is dangerous partly because of who drives then and how.** The night odds ratio shrinks from 1.98 to 1.48 once alcohol and speeding are included, but it does not disappear. Darkness, fatigue or other unmeasured night-time factors may also play a role.
- **Rural roads carry higher severity.** Rural crashes have about 40% higher odds of a severe outcome even after adjustment. This is consistent with higher speeds and longer emergency response times, which this analysis did not measure.
- **Bad weather means more crashes, not more severe crashes.** Rain and snow/ice are associated with lower odds of a severe outcome, most likely because drivers slow down. This is an interpretation of an association, not a tested mechanism.
- **Junctions are associated with lower severity than open road.** Intersection and driveway crashes are less severe than crashes along roadways, where speeds tend to be higher.

b. Stakeholders and implications

The findings can help several groups decide where to focus attention. Each stakeholder below is listed with the findings most relevant to them.

National agencies: NHTSA and the U.S. Department of Transportation.

NHTSA funds and coordinates national behavioral safety programs. Alcohol and speeding remain the strongest associations after adjustment, and they compound each other: 13.9% of crashes involving both are severe or fatal. This supports keeping impaired driving and speeding at the center of national campaigns. It also suggests that messages addressing the combination of the two may be especially valuable. The stability of these patterns across 2022–2024 indicates that they are persistent problems, not short-term fluctuations.

State departments of transportation and state highway safety offices.

States set annual targets for fatalities and serious injuries and write Strategic Highway Safety Plans. Rural crashes have about 40% higher odds of a severe outcome, and crashes away from junctions are more severe than those at intersections. These results point toward rural, non-intersection roadway segments as candidates for systemic safety treatments, such as rumble strips, shoulder improvements and speed management. Because this analysis did not include posted speed limits or roadway design, states would need their own roadway data to choose specific locations.

Local governments and metropolitan planning organizations (MPOs).

Many cities and regions have Vision Zero or local road safety plans. The time-of-day results show that crash volume and crash severity peak at different times. Volume peaks in the weekday afternoon commute, while severity peaks between midnight and 05:00, especially on weekend nights. Local plans that rank locations or programs only by crash counts may therefore underweight the conditions that produce the most serious harm.

The traveling public.

For drivers, the most direct message is that the combination of alcohol, speeding and night driving sharply increases the chance that a crash is serious. In the model's example, the predicted probability rises from about 2% for a typical daytime urban crash to about 22% when these factors occur together on a rural road.

c. Limitations

- CRSS is observational. Unmeasured factors such as impact speed, seat-belt use, vehicle type and EMS response may explain part of the associations.
- The project includes police-reported crashes only. Crashes never reported to police are not represented.
- Descriptive estimates use the CRSS weights. The regression uses normalized weights and PSU-clustered standard errors, which ignore the sampling strata and are flagged by statsmodels as not fully supported. A design-based check with strata and PSUs confirmed all main conclusions, but p-values should be treated as approximate, and borderline results (interchange/ramp, 2023 vs 2022) are not interpreted.
- Alcohol (imputed by NHTSA when unknown), speeding and injury severity depend on police reporting and judgment at the scene.
- Speeding comes from vehicle records aggregated to the crash.
- About 4% of crashes with unknown speeding or interstate status are not in the model, and they are somewhat more severe than average.
- The model does not include driver characteristics, vehicle type or posted speed limit. The low pseudo- R^2 (0.037) reflects how much of crash severity these few characteristics leave unexplained.
- The results describe 2022–2024 and should not automatically be generalized to other periods.

d. Conclusion

In U.S. police-reported crashes from 2022 to 2024, alcohol involvement (OR 2.77) and speeding (OR 2.00) are the characteristics most strongly associated with a severe or fatal outcome. They are followed by nighttime (1.48), a rural location (1.41) and weekends (1.26). Adverse weather is associated with less severe outcomes. Combined, the high-risk characteristics raise the predicted probability of a severe/fatal outcome from about 2% for a typical daytime urban crash to about 22%. The patterns are stable across the three years. Using the CRSS sampling weights was essential: without them, the severe/fatal share would be overstated fourfold. All conclusions are associations, not causal effects. For transportation-safety stakeholders, these results suggest that efforts to reduce deaths and serious injuries should focus not only on where crashes are most frequent, but also on the conditions that make crashes most severe, which include impaired and speeding drivers, nighttime and weekend travel, and rural roads.

References

- Federal Highway Administration. *Safety Performance Management Measures* (23 CFR Part 490, Subpart B).
- National Highway Traffic Safety Administration. *Crash Report Sampling System (CRSS)*, 2022, 2023 and 2024 data files. National Center for Statistics and Analysis, U.S. Department of Transportation. nhtsa.gov/crash-data-systems/crash-report-sampling-system
- National Highway Traffic Safety Administration. *CRSS Analytical User's Manual* and *CRSS Coding and Validation Manual*. National Center for Statistics and Analysis, U.S. Department of Transportation.
- Seabold, S., & Perktold, J. (2010). statsmodels: Econometric and statistical modeling with Python. *Proceedings of the 9th Python in Science Conference*.
- U.S. Department of Transportation. (2022). *National Roadway Safety Strategy*. Washington, DC: U.S. DOT.